



Sri Lanka Institute of Information Technology

Web-based Salon Management System

Project Report

Information Systems Project 2026

Project ID: G_19

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A photograph of a handwritten signature in dark ink on a light-colored background. The signature is written over a horizontal dotted line. The signature appears to be 'Hiruni Peiris'.

Ms. Hiruni Peiris

27/05/2026

Date of submission

Abstract

Salons of small and medium scale in Sri Lanka, like Harshi Salon, still use a manual procedure for scheduling and managing appointments and customers using books. This leads to various issues such as double booking, scheduling conflict, payment problems, no feedback system in place, and others. As a result, the efficiency of such businesses is affected negatively. For resolving those problems, a web-based Salon Management System was built specifically for Harshi Salon. This application allows scheduling, re-scheduling, and cancellation of appointments by users, manages the role-based user management, allows service/package management, makes payments, gives a structure for feedback and rating, and includes a personalized recommendation system for hairstyles. All manual processes of appointment booking were eliminated thanks to this solution, scheduling conflicts disappeared, and human errors while keeping records and generating reports were avoided. The testing of this system showed that it satisfies all of the functional and non-functional requirements set out earlier in this paper. The advantages of the new system include better efficiency, increased convenience for customers, the ability to make decisions based on data, and improved working conditions for salon workers.

Acknowledgement

We would like to sincerely thank all our project group members for their cooperation and teamwork throughout the Information Systems Project. We are also grateful to our client for providing valuable guidance and clear requirements that supported the development of this system. Finally, we extend our sincere appreciation to our lecturer and supervisor for their continuous guidance, feedback, and encouragement, which greatly contributed to the successful completion of this project.

Declaration

We declare that the this project report or part of it was not a copy of a document done by any organization, university any other institute or a previous student project group at SLIIT and was not copied from the Internet or other sources.

Project Details

Project Title	Web-based Salon Management System
Project ID	G_19

Group Members

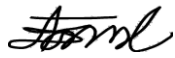
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1. Introduction

1.1 Problem Statement

In Sri Lanka, many small and medium-scale salons, including Harshi Salon, still rely on traditional appointment booking methods where customers are required to call or visit the salon to schedule their appointments. This manual approach highlights the need for a more efficient and structured digital solution to improve service delivery and overall customer experience.

For Harshi Salon, the current system presents several operational challenges. The reliance on a notebook-based system to maintain customer details and schedules leads to difficulties in managing appointments effectively. During peak hours, issues such as double bookings, appointment clashes, and time delays frequently occur. In addition, when phone lines are busy or unanswered, customers experience inconvenience when attempting to make bookings.

Furthermore, the manual system makes it difficult to maintain accurate records of customer information and payments. The absence of a structured feedback mechanism limits the salon's ability to evaluate customer satisfaction and service quality. Additionally, the reliance on manual processes increases the likelihood of human errors and inconsistencies during monthly report generation. This affects the accuracy of business insights and makes it challenging for the owner to make informed decisions to improve overall service efficiency.

1.2 Product Scope

The system being proposed is the Salon Appointment Management System. This system is accessed over the web and is to be used by Harshi Salon. The idea behind this system is to replace the current manual, paper based booking system with a streamlined digital booking system.

The system will cater for both customers, salon staff and the owner. The system was designed to increase operational efficiency through reduction of errors due to manual operation, avoiding and providing valid reasons for appointment conflicts, and by presenting all the necessary information

accurately and clearly. This allows the system to facilitate communication between customers and the salon.

1.2.1 Benefits

The Salon Management System is supposed to help both the salon and its customers. It has a feature that recommends styles based on customer preferences, which helps customers identify suitable hairstyles or services, improving decision-making and enhancing the overall customer experience. The system enables customers to book appointments online, eliminating the need for phone calls or physical visits, which improves convenience and reduces delays in the booking process. In addition, the system allows customers to reschedule or cancel appointments easily, providing greater flexibility in managing their bookings and reducing scheduling conflicts for the salon. The system improves efficiency, enhances customer satisfaction, and modernizes salon operations by replacing manual processes with a structured digital solution, making things more efficient and bringing the salon up to date.

1.2.2 Objectives

1.2.2.1 Main Objective

To develop and deploy a web-based Salon Management System for Harshi Salon within the project development period that enables appointment booking, customer and service management, package handling, payment processing, feedback collection, and style recommendations, while improving operational efficiency, reducing manual errors, and achieving improved user satisfaction.

1.2.2.2 Sub Objectives

1. Admin and Customer Management

To ensure efficient and secure management of customer and admin information within the project duration by enabling accurate creation, updating, and retrieval of user data, thereby improving data consistency and system reliability.

2. Service Management

To improve the organization and accessibility of salon services within the system by enabling structured management of service details, ensuring accurate and up-to-date service information for users.

3. Package Management

To enhance service flexibility within the salon system by enabling the creation and management of service packages that combine multiple services, improving customer choice and service value.

4. Appointment and Payment Management

To improve booking efficiency and reduce scheduling conflicts by enabling online appointment booking, rescheduling, cancellation, and accurate tracking of payment transactions within the system.

5. Feedback and Rating System

To improve service quality and customer satisfaction by enabling customers to provide structured feedback and ratings, allowing the salon to evaluate and enhance service performance.

6. Style Recommendation System

To enhance customer decision-making and personalization by providing automated hairstyle and service recommendations based on user preferences within the system.

1.3 Project Report Structure

This section provides an overview of the structure of the project report and briefly explains the content of each chapter to guide the reader through the document.

Requirements and Analysis

The chapter highlights the requirements of the proposed system both functional and nonfunctional along with user needs and constraints that informed the system's development.

Design

This section details the architecture of the system along with database/file structure and UML diagrams of system components interaction.

Implementation

It focuses on the implementation of the system using chosen technologies and explains the implementation of key modules of the system.

Testing

The chapter describes the testing methodology for the system verification that includes testing scenarios and their outcomes to ensure system requirements compliance.

Assessment of the Project Results

The chapter assesses the developed system based on the analysis of its performance compared to initial requirements and expectations.

Lessons Learned

This chapter describes lessons learned from the course of system development, including improving skills in technical and problem-solving areas.

Future Work

It gives an outlook on future improvements of the system and possible new functionalities that can be included in the next iterations.

2. Methodology

2.1 Requirements and Analysis

Salon Management System development starts with requirements elicitation. This phase pretty is crucial because it helps identify the system's functional and non-functional requirements. The goal is to create a final product that meets all stakeholders' needs.

First, the stakeholders were identified and grouped into primary, secondary, and tertiary stakeholders. This helps understand user roles and how they interact with the system. It also aids in prioritizing requirements based on each stakeholder group's importance and influence on system functionalities.

Requirements Elicitation Techniques

To gather system requirements, multiple elicitation techniques were used, including interviews, questionnaires, and observation.

Interviews

The salon owner was interviewed to understand the challenges in managing appointments, customers, and services. This led to key system requirements like appointment scheduling and customer management.

Questionnaires

Questionnaires were sent to other stakeholders to collect feedback. The responses showed what users expect, the difficulties with the current manual system, and the desired features.

Observations

Observing the salon's daily operations helped identify practical issues. For example, scheduling conflicts and delays in booking were common problems.

Derivation of System Requirements

After analyzing the gathered information, the system requirements were defined. The key features include:

1. Appointment management
2. Customer and user management
3. Service and package management
4. Payment handling

5. Feedback and rating system

6. Style recommendation system

These requirements are the foundation of the system design and implementation. They ensure the system meets the stakeholders' needs and is functional.

2.1.1 Functional Requirements

Epic - Administration and User Management

FR1: The system shall permit users to register a user account.

FR2: The system shall permit users to login/logout securely.

FR3: The system shall permit users to manage their user profiles.

FR4: The system shall allow role-based access to administrator, staff and customers.

FR5: The system shall permit administrators to manage user accounts.

FR6: The system shall permit users to change password.

FR7: The system shall permit viewing user's customer service history.

Epic - Services Management

FR8: The system shall allow administrators to manage salon services such as add, edit, and delete.

FR9: The system shall allow salon services management.

FR10: The system shall allow users to view salon services according to categories.

FR11: The system shall allow users to view the details about each of the salon services.

Epic - Packages Management

FR12: The system shall allow administrators to manage service packages such as add, edit, and delete.

FR13: The system shall allow users to view/browse service packages.

FR14: The system shall allow users to view details about service packages.

Epic – Feedback and Rating System

FR19: The system should enable the submission, editing, and deletion of customer feedback.

FR20: The system should facilitate administrators' viewing of ratings and reviews.

Epic – Style Recommendation System

FR21: The system should provide the functionality for administrators to configure the rules for the style recommendation algorithm.

FR22: The system should make personalized style recommendations for the customers.

2.1.2 Non-Functional Requirements

NFR1 – Performance: The system should ensure loading times of no more than 3 seconds for 95% of page loads in normal conditions with up to 100 simultaneous users.

NFR2 – Usability: The system should enable users to perform basic tasks (registering, booking, paying, submitting feedback) in 3-5 steps without any technical skills.

NFR3 – Security: The system should protect password data by using encryption, as well as enforce role-based access restrictions.

NFR4 – Reliability: The system should operate 99% of time in operational mode through efficient data management and backup procedures.

NFR5 – Scalability: The system should be capable of future growth in terms of numbers of users, services offered, and operations performed.

2.1.3 Use Case Diagram

The diagram highlights the key actors – Customer, Owner, Receptionist, and Staff, and their role within the system. As seen from the diagram, customers have the ability to schedule appointments, make payment, offer feedback, and receive style advice, whereas owner and receptionist handle administrative duties related to users, services, and packages. Staff will be able to view their assigned appointments and service feedback from customers.

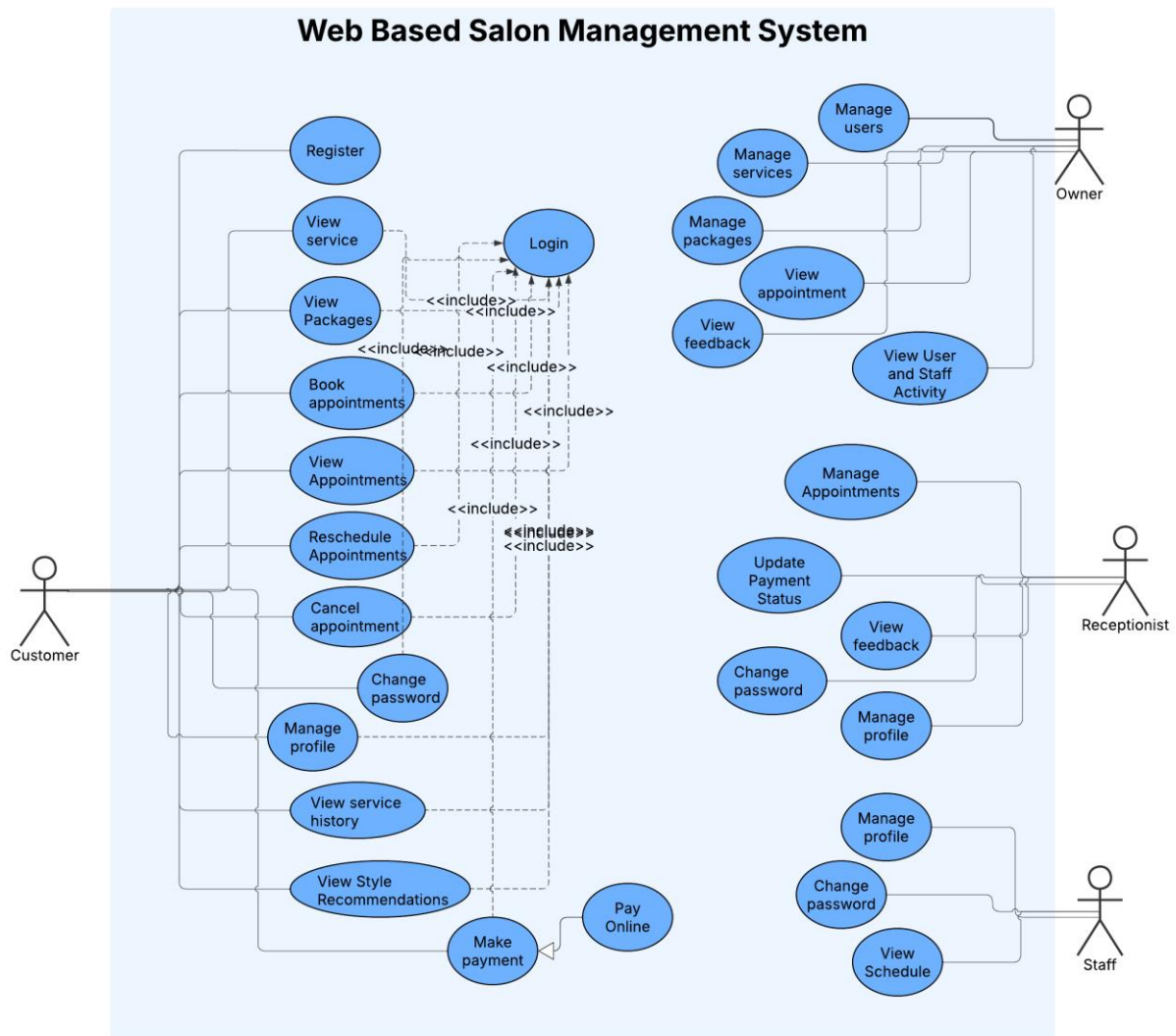


Figure 2.1 : Use Case Diagram

2.1.4 Activity Diagrams

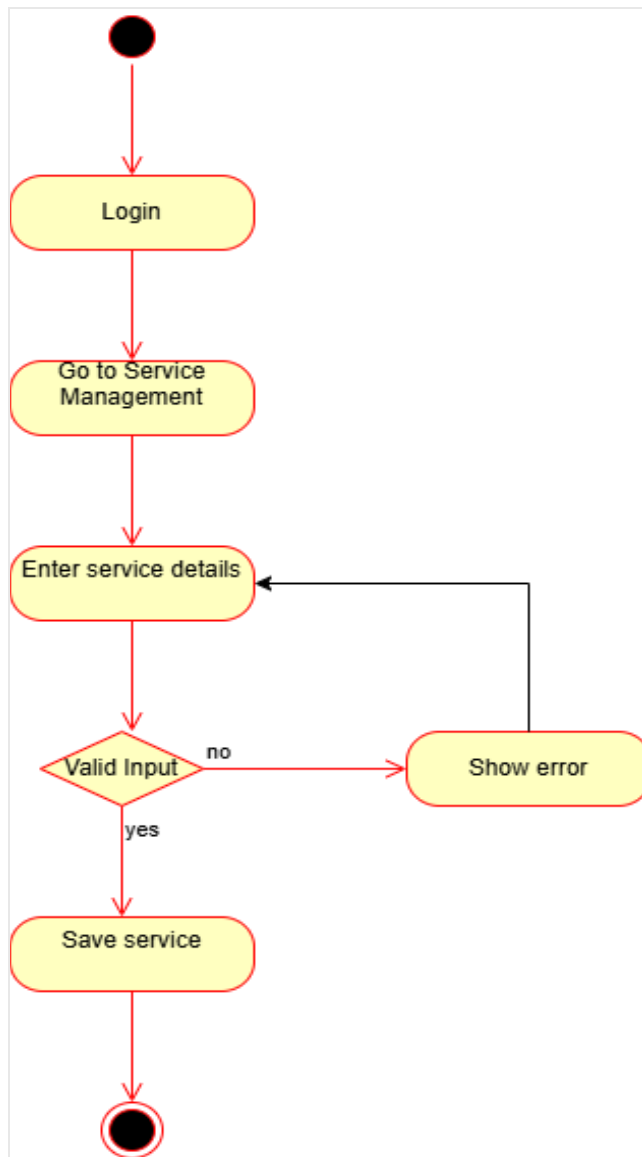


Figure 2.2 : Add Service Activity Diagram

The owner logs in and navigates to the Service Management section. They enter the details of the new service. If the input is invalid, an error is displayed and the owner is prompted to re-enter the details. Once valid input is confirmed, the service is saved to the system and becomes available for customers to view and book.

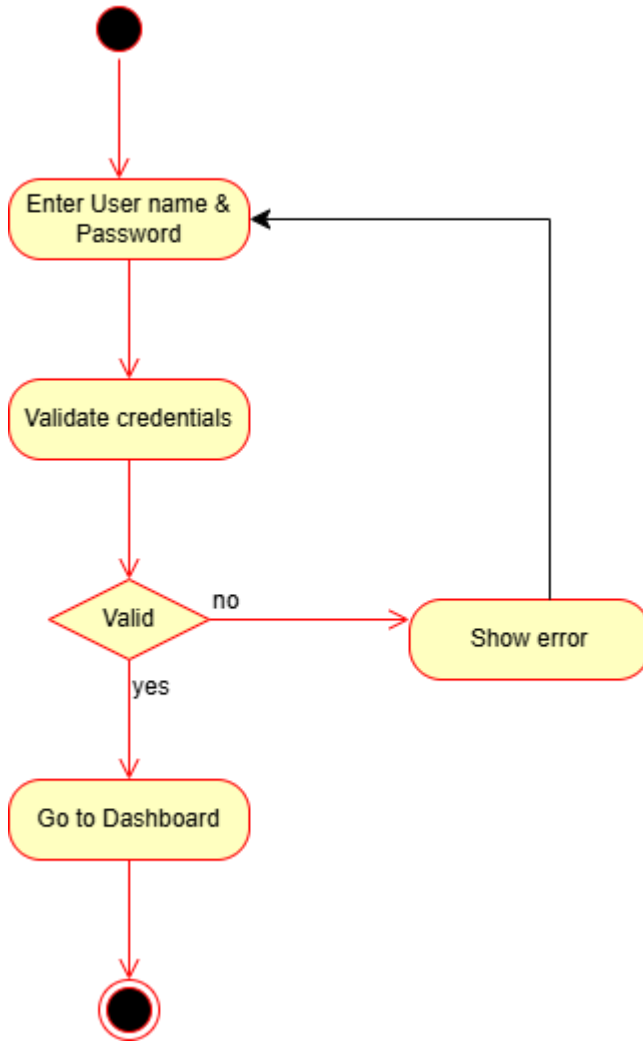


Figure 2.3 : User Login Activity Diagram

The login process begins when the user enters their username and password. The system validates the credentials — if invalid, an error message is displayed, and the user is prompted to re-enter their details. Upon successful validation, the user is redirected to their dashboard. This applies to all actors: Customer, Receptionist, and Staff.

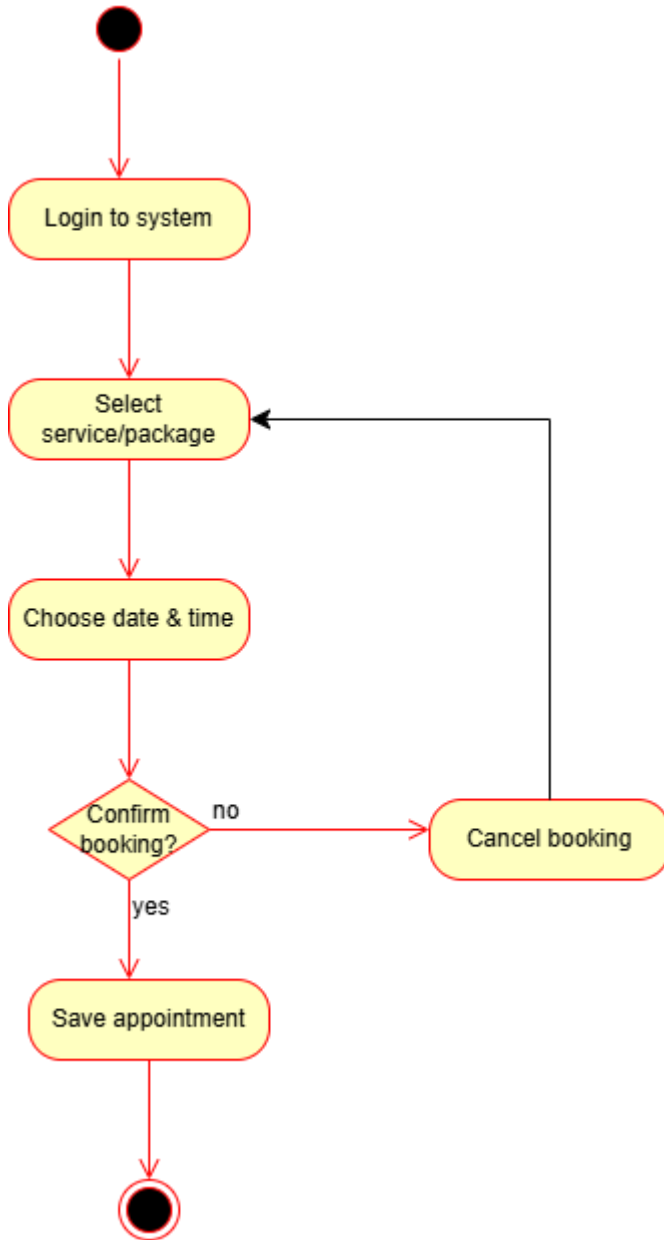


Figure 2.4 : Book Appointment Activity Diagram

After logging in, the customer selects a desired service or package. They then choose a preferred date and time. If the customer confirms the booking, the appointment is saved to the system. If not confirmed, the booking is cancelled and the customer may select a different service or package.

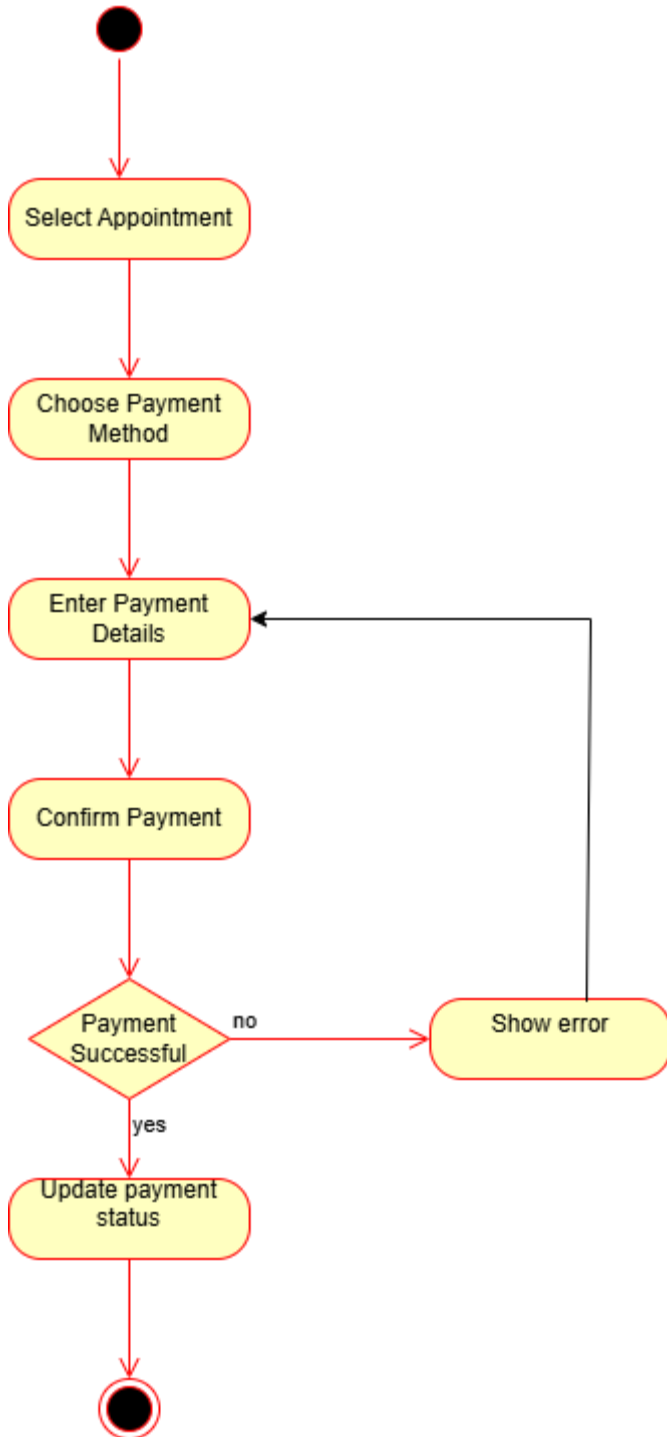


Figure 2.5 : Make Payment Activity Diagram

The customer selects an appointment to pay for and chooses a payment method. They then enter the required payment details. If the payment fails, an error is shown and the customer is returned to re-enter their details. On successful payment, the system automatically updates the payment status for the receptionist to view.

2.2 Design

2.2.1 ER Diagram

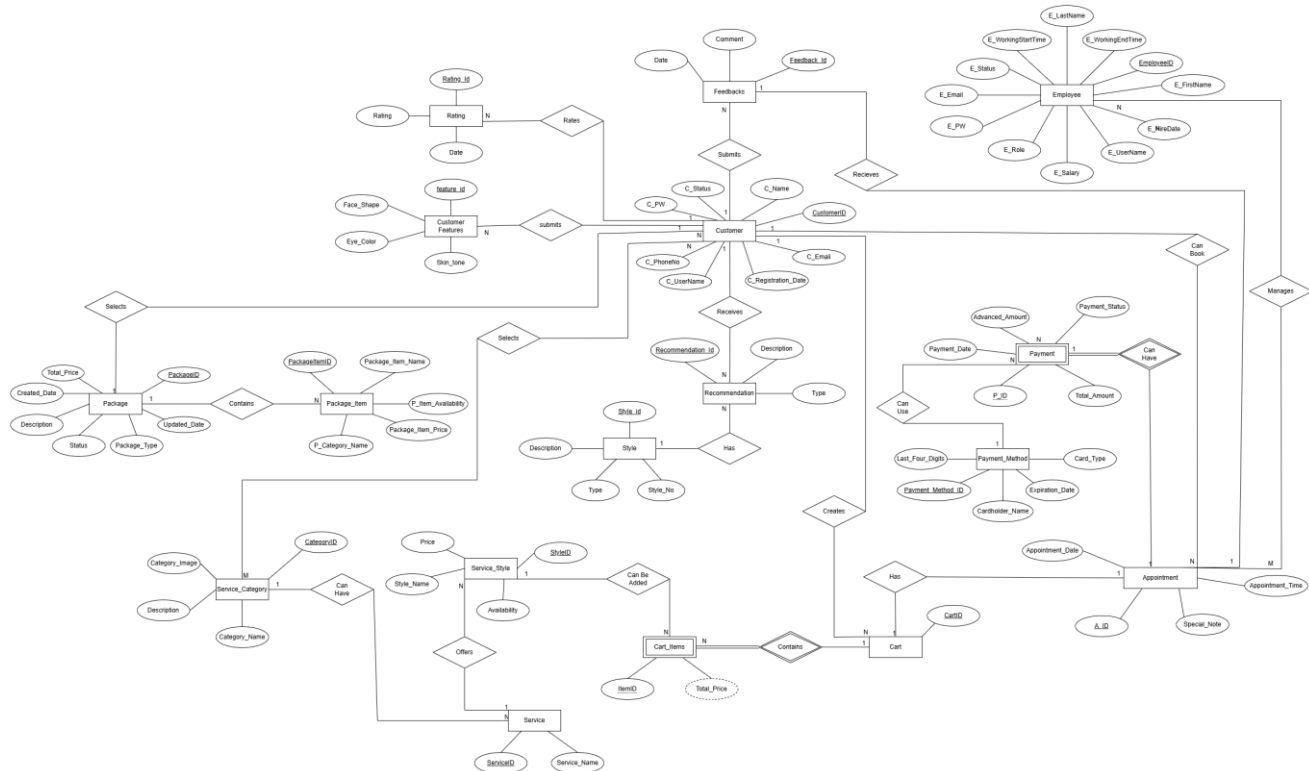


Figure 2.6 : ER Diagram

The entity-relationship (ER) diagram depicts the structure of the database and relationship between entities for the web-based salon management system. This system operates on the Customer entity that enables clients to make bookings, perform payment operations, give feedbacks and rate services. At the same time, customers will be offered hairstyles in accordance with features that they have. Appointments are handled by the employees using the Cart system in which customer picks services to be used in their appointments. The service management module consists of such components as Service Category, Service, and Service Style, which helps customers pick appropriate services according to their needs. There are also components like Packages and Package Items that provide clients with packages of services; and Payments and Payment Methods that manage transactions between customers and the company. Furthermore, there is the recommendation module that uses Customer Features, Styles and Recommendations to generate appropriate styles.

2.2.2 Logical Schema

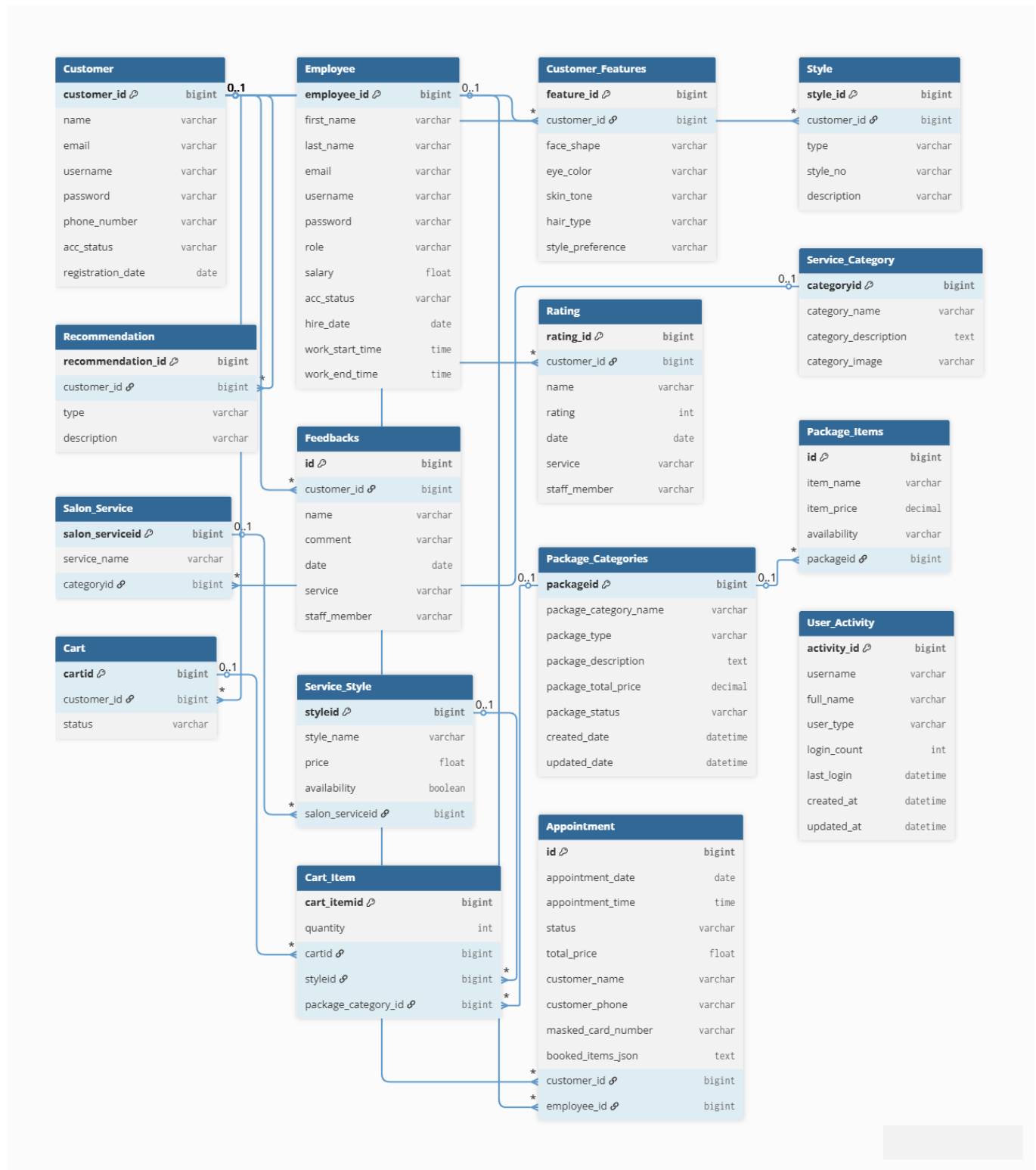


Figure 2.7 : Logical Schema

A logical database schema will consist of entities such as Customer, Appointment, Feedbacks, Rating, Recommendation, Cart, and Style among others, each having their attributes, keys, and relations. The Customer entity is an important part of the Web-based Salon Management System and relates to other entities such as Cart, Appointment, Feedbacks, Rating, Recommendation, Customer_Features, and Style. Employee data will relate to Appointment, whereby employees will be responsible for dealing with appointments.

Management of services within the system will include Service_Category, Salon_Service, and Service_Style. This implies that services will be categorized and will be available in specific styles. Cart and Cart_Item will take care of the customer selection prior to the booking, taking into consideration service styles and packages. Entities such as Package_Categories and Package_Items will take care of packages and will relate to Appointment.

Recommendation, User Activity, and Customer_Features will manage the provision of style recommendations and user activities within the web-based system.

2.2.3 Class Diagram

Class diagram depicts the static view of the structure of the system, as well as the classes, their attributes, methods, and their relationships. The class diagram shows the interaction between various objects in the Salon Management System.

User is the parent class in which customer, staff, receptionist, and owner share common attributes like name, email, and password along with their own unique methods.

Multiplicity values are used to depict the relationship between classes; the multiplicity values represent the number of occurrences. An example of this would be that 1 customer is related to more than 1 appointment.

Aggregation relationship between the class "Service" and "Service Package" depicts that service packages are made up of several services aggregated together into a single package.

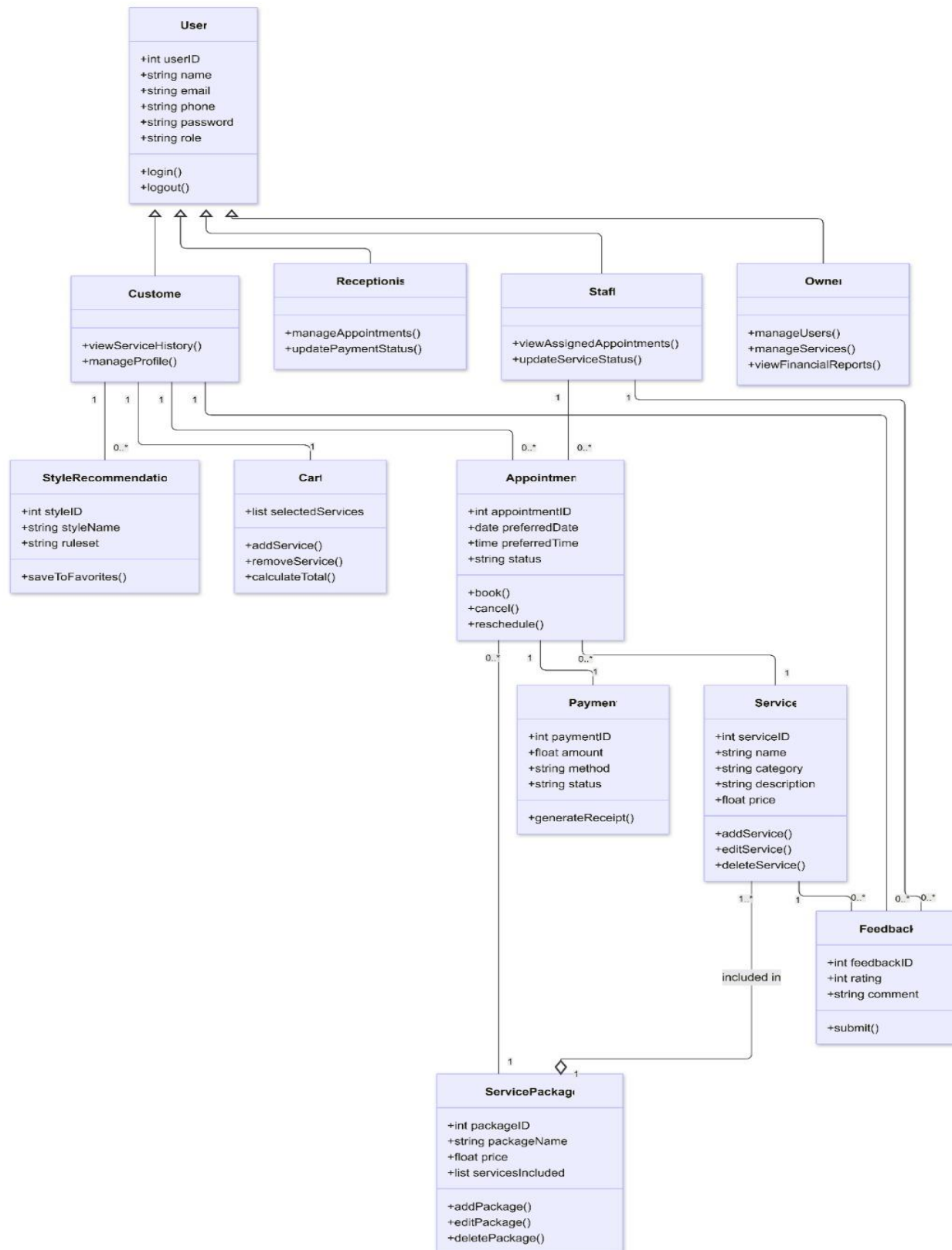


Figure 2.8 : Class Diagram

2.2.4 Component Diagram

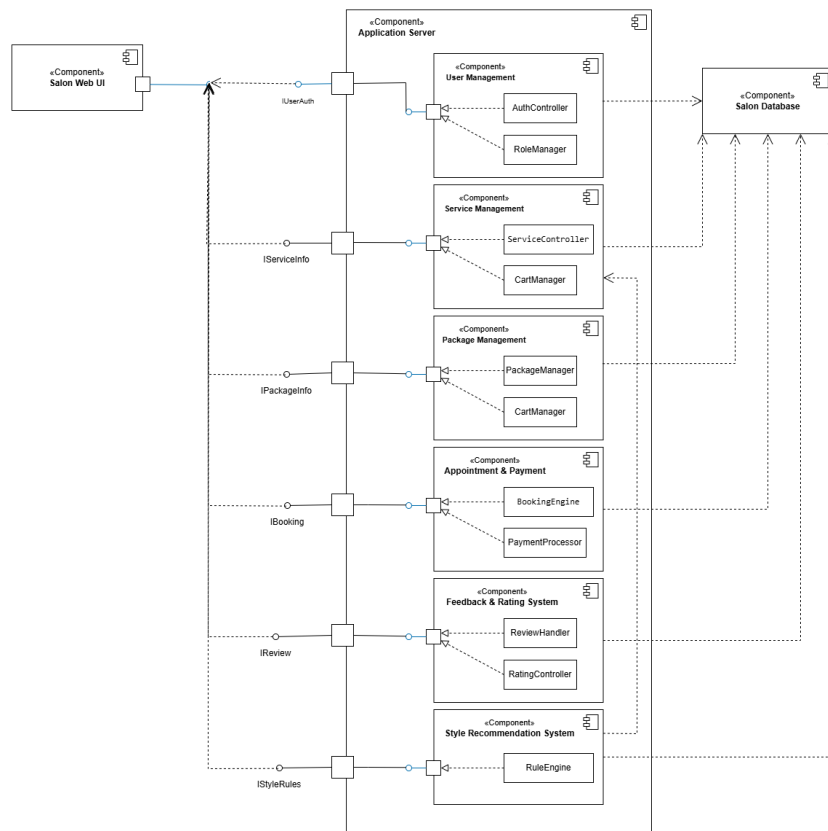


Figure 2.9 : Component Diagram

As illustrated in the component diagram, the component diagram is an illustration of the physical structure of the system, and also illustrates how different components work together to form a cohesive unit. There are three key layers in the system: Presentation Layer, Business Logic Layer, and Data Layer. These layers perform specific tasks to ensure that there is separation of concerns.

2.2.5 Deployment Diagram

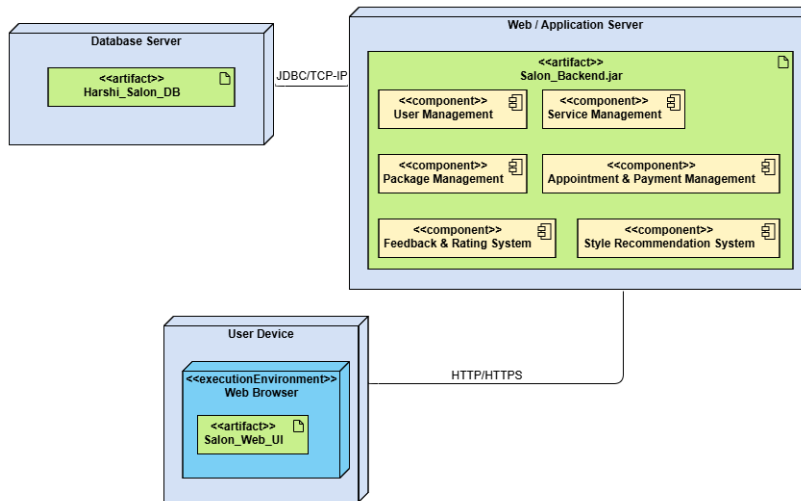


Figure 2.10 : Deployment Diagram

The deployment diagram illustrates the physical deployment of the Salon Management System and how the software components are distributed across hardware nodes. It consists of three main nodes: the user device, the web/application server, and the database server.

The user interacts with the system through a web browser on the user device, which accesses the application via the internet. The application server hosts the system's backend logic, while the database server is responsible for storing and managing all system data.

The diagram also shows how the system components communicate across these nodes to ensure proper execution and data exchange within the system environment.

2.2.6 State Diagram

The state diagram explains the various states that the Feedback and Rating process goes through when running within the system for Salon Management. The process starts in idle state, and it can move to states for submitting feedback or ratings depending on the action taken by the users. The users can also edit or undo any of the submissions done by them.

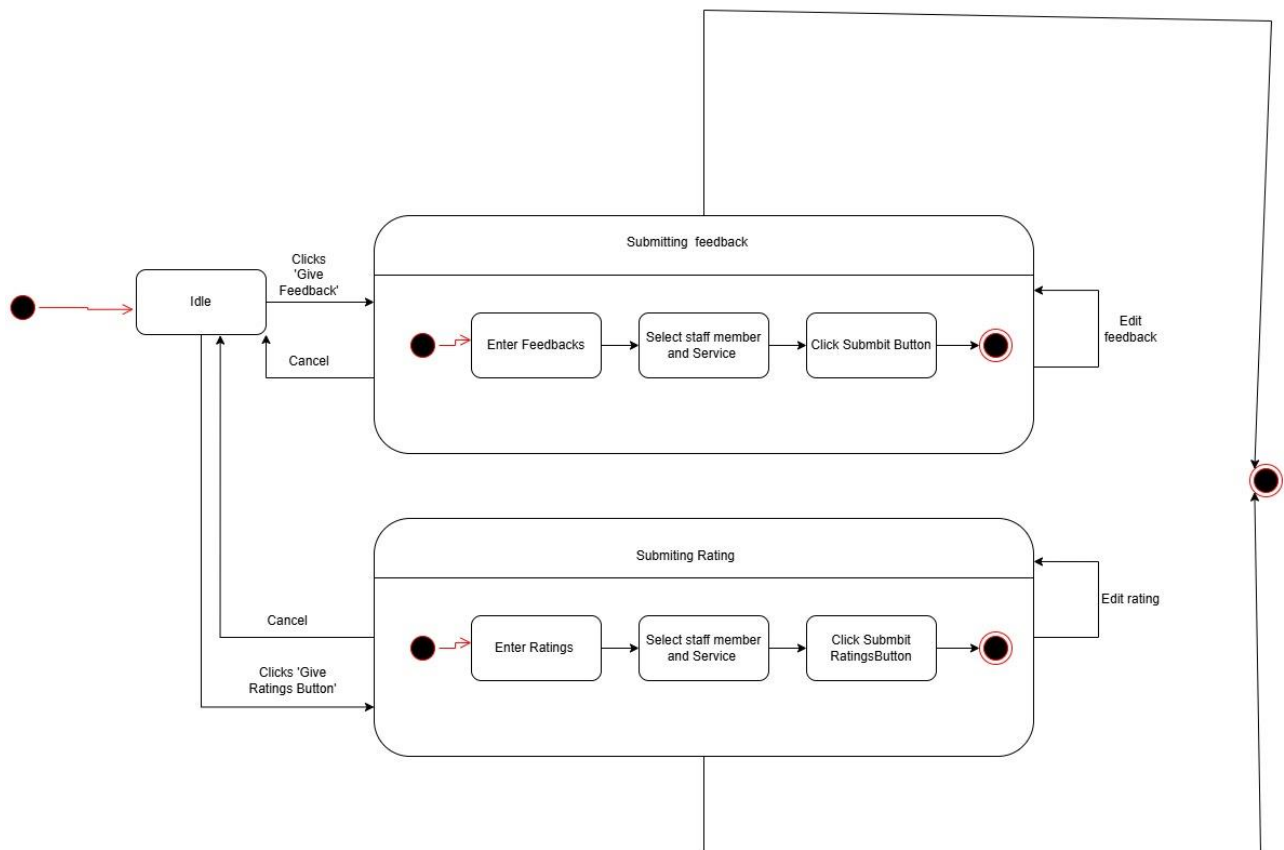


Figure 2.11 : State Diagram

2.2.7 Sequence Diagrams

2.2.7.1 Customer and a rule-based Style Recommendation System Interaction

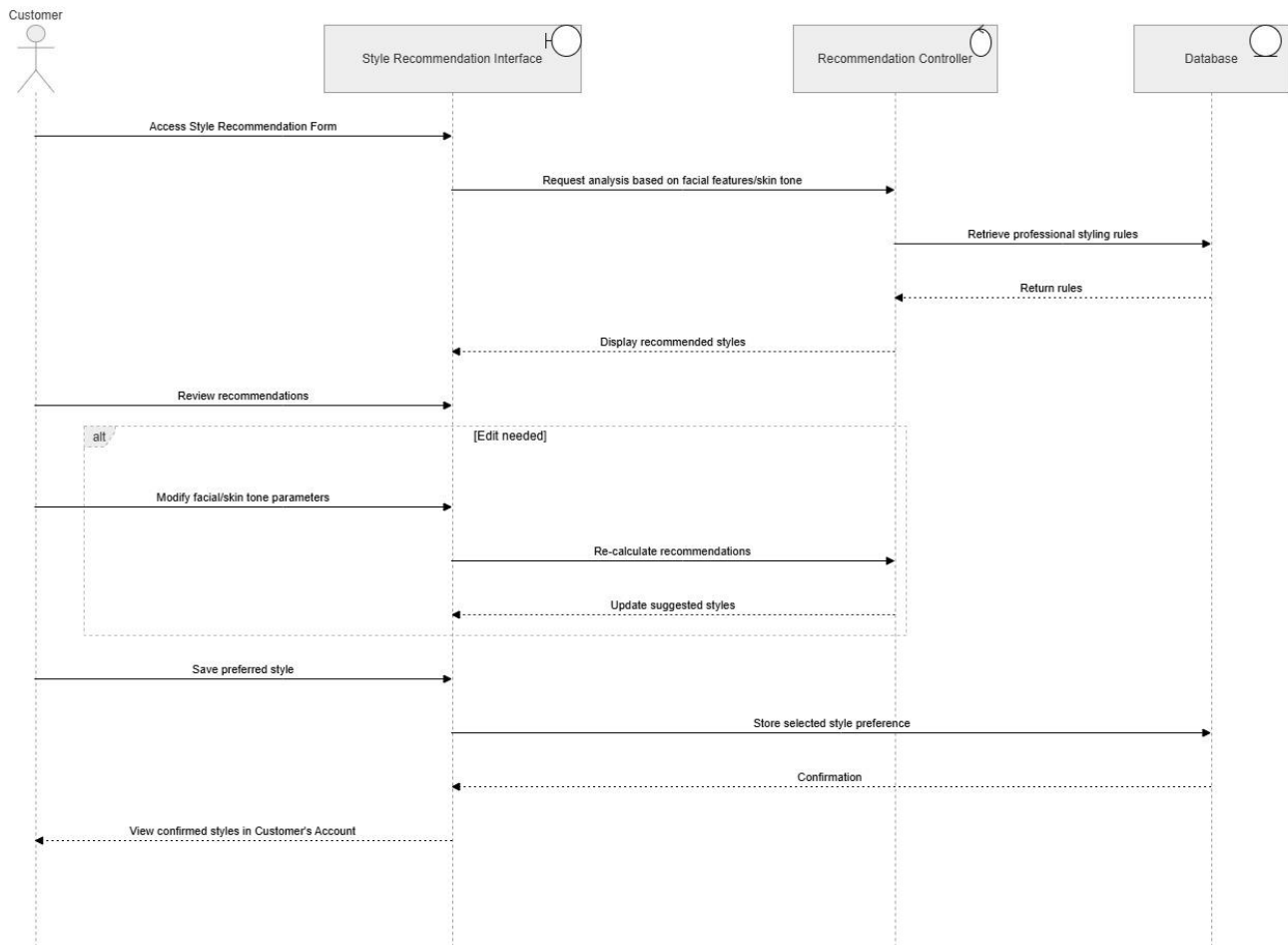


Figure 2.12 : Sequence diagram of customer and a rule-based Style Recommendation System Interaction

This sequence diagram represents the interaction of the customer and rule-based style recommendation system. The interaction explains how the input of the customer is used according to the defined styling rules for making recommendations.

2.2.7.2 Procedure of Customer Submits Feedback

The following diagram depicts the process of customer submits feedback. This depicts how customer feedback is captured and stored in the database and how it is modified.

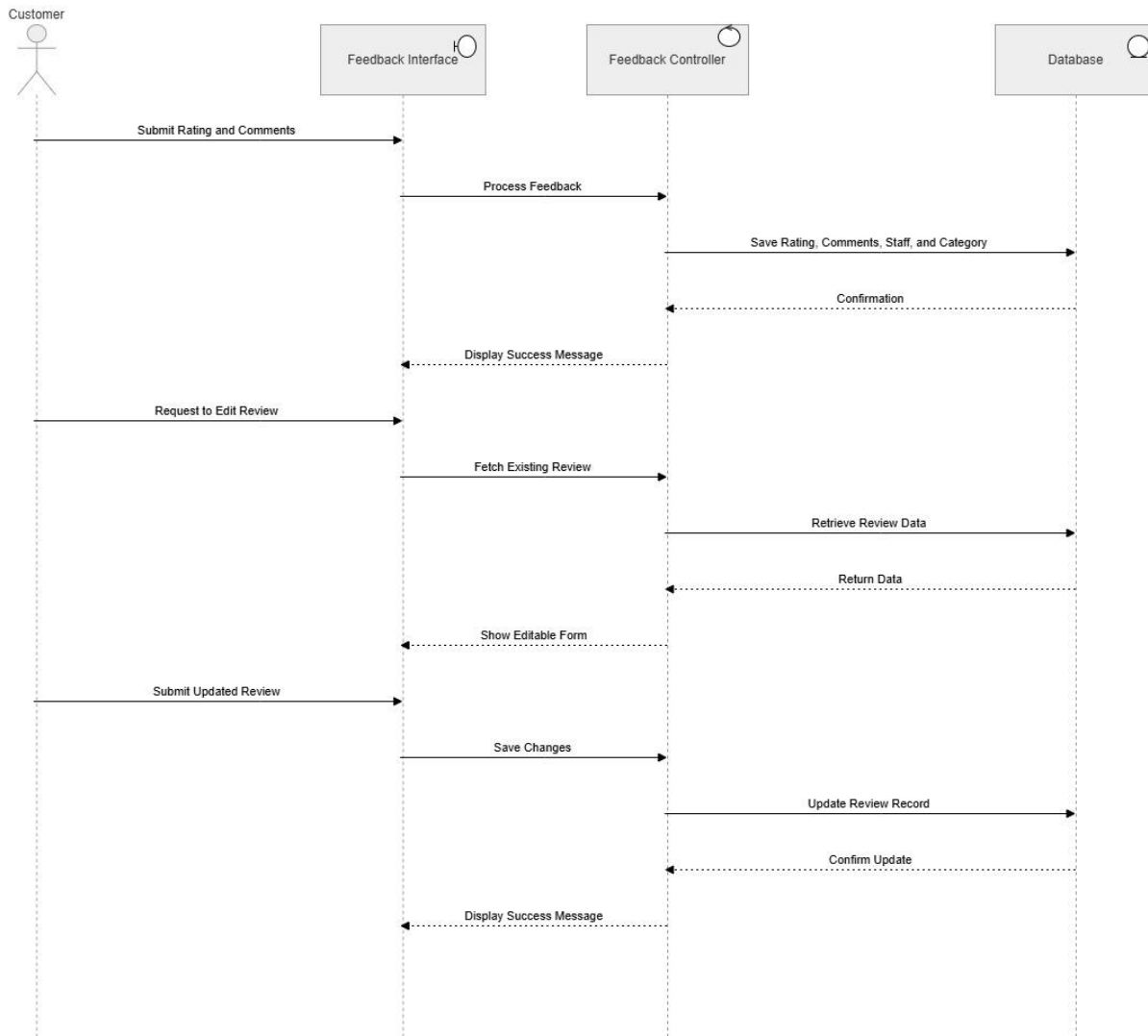


Figure 2.13 Sequence diagram of customer submits feedback

2.2.7.3 Admin Managing a Package

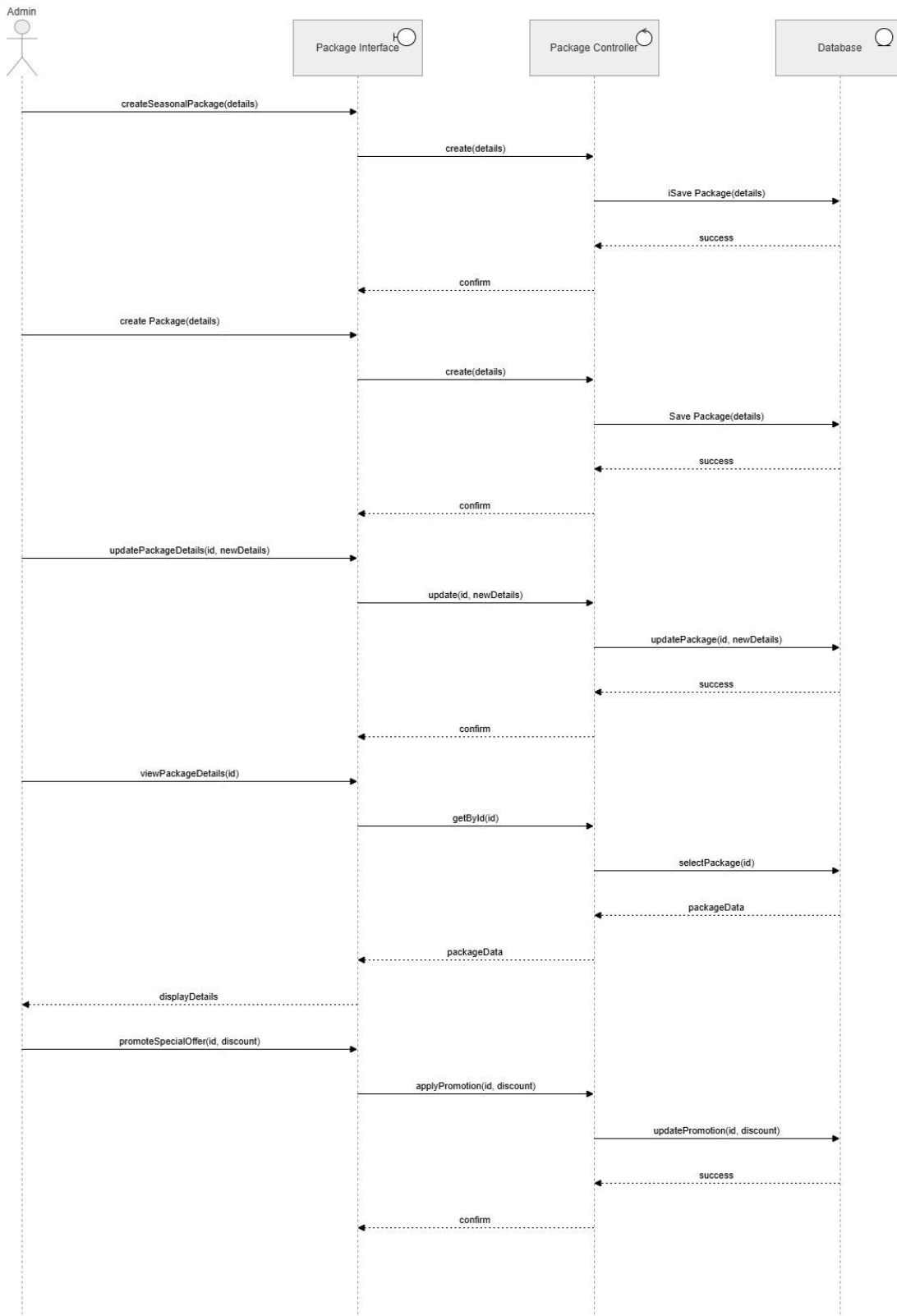


Figure 2.14 : Admin Managing a Package

The above interaction sequence diagram represents package management operations. The diagram represents how an administrator creates, updates, views, and manages packages from the salon and stores the information in the database.

2.2.7.4 User Management Operations

The following interaction sequence diagram represents user management operations such as user registration, user login, user modification, and user deletion.

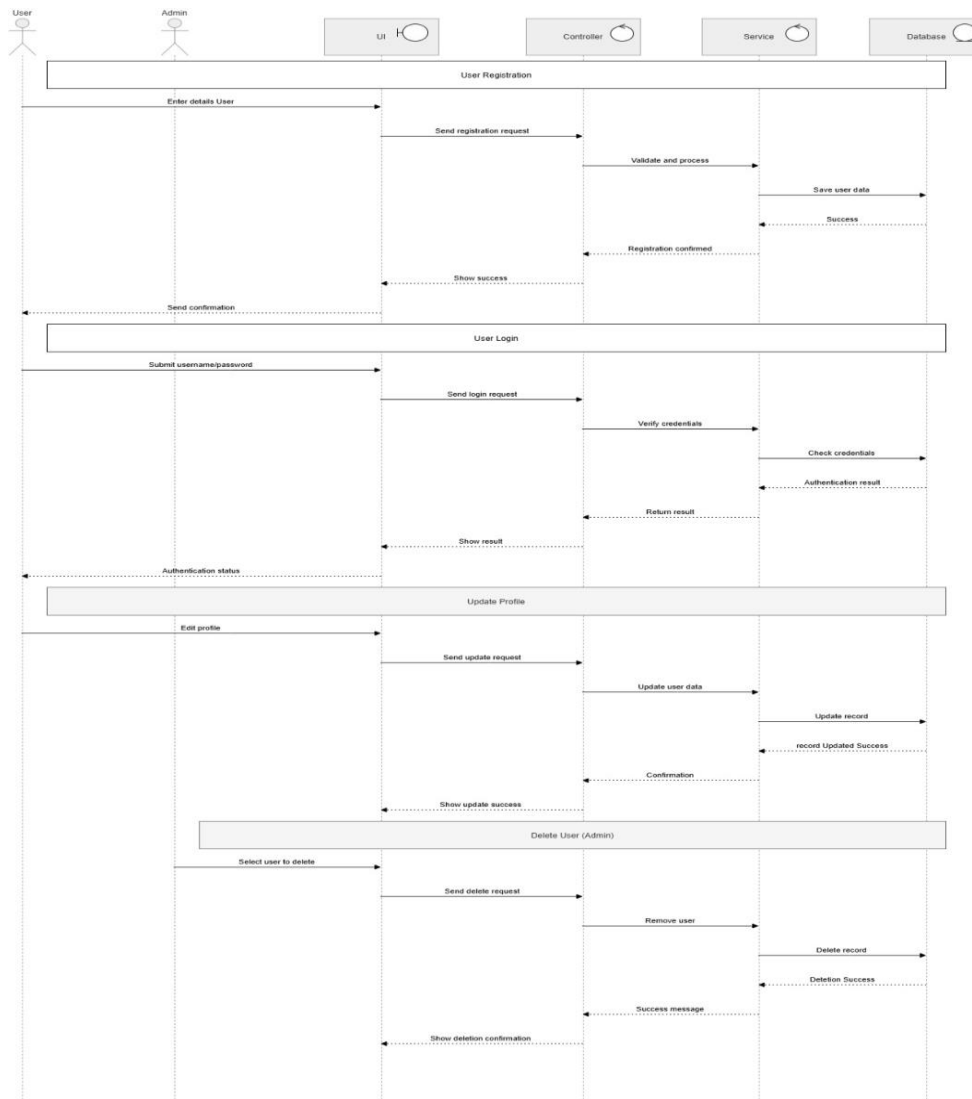


Figure 2.15 : User Management Operations

2.3 Implementation

2.3.1 Implementation Overview

The Salon Harshi Management System is a complete web application built based on a multi-layered MVC (Model View Controller) framework. This application is based on the Spring Boot framework that consists of server-side rendering for the frontend alongside a REST API for the backend, which connects to Microsoft SQL Server. It provides functionalities related to six modules – Admin and Customer Management, Services Management, Packages Management, Appointments and Payment Management, Style Recommendation System, and Feedback & Ratings Management.

2.3.2 Major Module Implementation

2.3.2.1 Admin and Customer Management

This component handles user authentication and authorization for users such as Owners, Receptionists, Staff, and Customers. It makes use of Spring Security to handle login procedures, password encoding, roles and authorizations, and dashboard redirection according to user role.

2.3.2.2 Service and Package Management

This component allows for the creation, modification, viewing, and management of salon services and packages. Services can be organized into categories, while packages include a combination of services and allow customers to get service bundles conveniently.

2.3.2.3 Appointment and Payment Management

This component lets customers make appointment bookings, schedule rescheduling and cancellation of existing appointments. It allows for efficient scheduling management for both staff and receptionist employees. Payment processing also takes place within this component.

2.3.2.4 Style Recommendation System

This component is designed to generate customized hairstyle recommendations for clients based on their face shapes, skin tones, hair types, and styles among other factors. The style recommendation system employs a rules-based algorithm for generating recommendations.

2.3.2.5 Feedback and Rating System

This component allows for the submission of ratings and feedback for salon services. The feedbacks are recorded in the database to help management review how well customers appreciate the provided salon services.

2.3.3 Implementation Languages and Tools

Component	Technology
Backend Language	Java 17
Application Framework	Spring Boot 3.3.3
ORM / Data Access	Spring Data JPA (Hibernate)
Security	Spring Security 6
Frontend Templating	Thymeleaf 3 + Thymeleaf-Security extras
Frontend Scripting	Vanilla JavaScript (Fetch API)
Styling	Custom CSS (CSS variables, no framework)
Database	Microsoft SQL Server (mssql-jdbc driver)
Build Tool	Apache Maven
Development Tool	IntelliJ IDEA (Spring Boot DevTools for hot reload)
Input Validation	Jakarta Bean Validation (@Valid, @NotNull, regex)

Table 2.1 : Languages and Tools used summary

2.3.4 Special Algorithm

The Style Recommendation System uses a rule-based recommendation algorithm that evaluates customer features and preferences to generate personalised styling suggestions. The detailed implementation of this algorithm is provided in Appendix C.

2.3.5 Reusable Components

Various reusable components have been created to enhance maintainability and avoid unnecessary code replication. Such components include services for authentications, validators, data transfer objects, user interfaces fragments, and global exception handling systems.

2.3.6 Choice of DBMS

The database management system chosen is Microsoft SQL Server, mainly due to its reliability, security, scalability, and compatibility with Spring Data JPA. The DBMS handles all the information about customers, appointments, payments, services, and feedbacks effectively.

2.4 Testing

2.4.1 Test Plan

Tests were done to ensure that all the functional requirements for the Salon Management System had been fully met and were working properly. Functional tests were executed on the modules such as Admin and Customer Management Module, Service Management Module, Package Management Module, Appointment & Payment Management Module, Feedback & Rating Management Module, and Style Recommendation System.

Test scenarios were created depending on the functional requirements obtained from the analysis phase. Test scenario contains preconditions, dependencies, execution steps, and the expected outcome of executing these steps. The aim of performing functional tests is to ensure that all features work as required by the functional requirement.

2.4.2 Test Cases and Results

2.4.2.1 Admin and Customer Management (E01)

Test Case ID: TC_UM_001	Test Designed by: Kahadawala D.C
Test Title: Customer Account Registration	Test Designed Date:
Test Priority: High	Test Executed by: Heiyanthuduwa D D M P
Module Name: User Management	Test Executed Date:
Description: Verify that a customer can successfully create a new account using their basic details so they can manage appointments.	
Preconditions: The user has a valid, unregistered email address and is on the registration page.	
Dependencies: Database connection must be active to store user credentials.	
Test Steps:	

1. Navigate to the Registration Page.
2. Enter a valid First Name, Last Name, and Contact Number.
3. Enter a valid Email address.
4. Enter and confirm a strong Password.
5. Click the 'Sign Up' button.

Post Conditions: The system creates the customer account, logs the data accurately, and redirects the user to the Login page.

Table 2.2 : Customer account creation test

	customer_id	acc_status	email	name	password	phone_number	registration_date	username
1	1	ACTIVE	kavi.d@gmail.com	Kavini DeSilva	\$2a\$10\$KY4a/KuX3B2TLzkVhpe0Oqp5BgIfb6cwzKBwlomoqp...	0701122335	2026-04-24	kavini
2	2	ACTIVE	ishara.fernando@gmail.com	Ishara Fernando	\$2a\$10\$8joMDAwVILEK5bJbvl.XOQF3WwXJ.FHILXllicca/S8wk...	0754567891	2026-04-24	ishara
3	3	ACTIVE	nethmi.perera@gmail.com	Nethmi Perera	\$2a\$10\$lwdiKNnrsbqQwdLCih.McOvW0UH9f.RvBvSH4Xwdu7...	0712345678	2026-04-24	nethmi
4	4	ACTIVE	dilmi.j@gmail.com	Dilmi Jayasinghe	\$2a\$10\$y104cx67nKvqLIPMwJIKWe3M/BK9vqvov/mHk9uKJu/X...	0785678912	2026-04-24	dilmi
5	5	ACTIVE	tharushi.a@gmail.com	Tharushi Anderson	\$2a\$10\$YvwV4JT3goQ7W2/39pxEOXyCaQdP6Y1.YB1DIW5Q...	0766789123	2026-04-24	tharushi

Figure 2.16 : Proof Accounts created successfully

2.4.2.2 Salon Service Management (E02)

Test Case ID: TC_SM_001	Test Designed by: Heiyanthuduwa D D M P
Test Title: Add Selected Service to Cart	Test Designed Date:
Test Priority: Medium	Test Executed by: Ranasinghe R A D S
Module Name: Service Management	Test Executed Date:
Description: Verify that a logged-in customer can add specific services to their cart to easily book an appointment.	
Preconditions: The customer is logged into the system and is browsing the service categories.	
Dependencies: Service categories and lists must be pre-populated by the admin.	
Test Steps: 1. Navigate to the Service Catalog. 2. Select a specific category (e.g., Hair Care, Nail Care). 3. View the service details (prices, descriptions). 4. Click the 'Add to Cart' button next to the desired service. 5. Open the Cart view.	
Post Conditions: The cart updates to display the newly added service along with the correct price.	

Table 2.3 : Testing Customer can add a Service



Figure 2.17 : Customer can successfully add service to book appointments

2.4.2.3 Package Management (E03)

Test Case ID: TC_PM_001	Test Designed by: Saraniya V.K
Test Title: Browse and Add Seasonal Packages	Test Designed Date:
Test Priority: High	Test Executed by: Heiyanthuduwa D D M P
Module Name: Package Management	Test Executed Date:
Description: Verify that a customer can view seasonal package details and seamlessly add the bundled package to their cart.	
Preconditions: The admin has created active seasonal packages	
Dependencies: The Package Management module is successfully integrated with the Service Management and Cart functions	
Test Steps: 1. Navigate to the Package section. 2. Select 'Seasonal Packages'. 3. Click on a specific package to view the services and details included. 4. Click the 'Add to Cart' button.	
Post Conditions: The selected package is added to the cart, accurately reflecting the bundled price rather than individual service prices.	

Table 2.4 : Testing Customer can add a package to cart

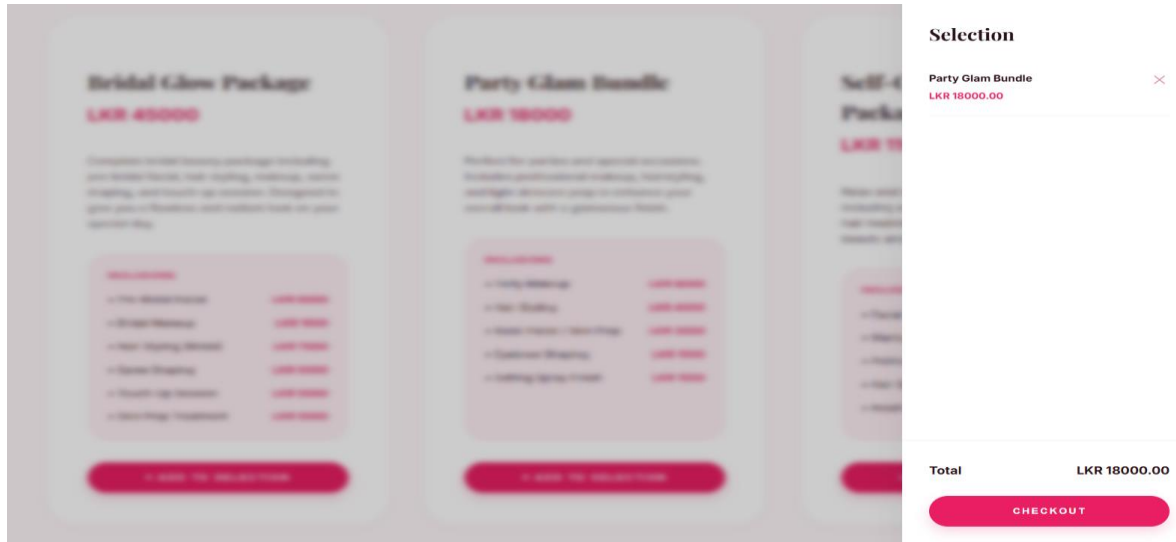


Figure 2.18 : Customer successfully added a package to cart

2.4.2.4 Appointment and Payment Management (E04)

Test Case ID: TC_APM_001	Test Designed by: Ranasinghe R A D S
Test Title: Reschedule Appointment via Calendar Interface	Test Designed Date:
Test Priority: High	Test Executed by: Heiyanthuduwa D D M P
Module Name: Appointment and Payment Management	Test Executed Date:
Description: Verify that a customer can successfully update or reschedule an existing appointment using the visual calendar interface.	
Preconditions (if there are any): The customer is logged in and has at least one upcoming appointment booked.	
Dependencies (if there are any): Calendar interface must be synced with the salon's available time slots.	
Test Steps: 1. Navigate to the 'My Appointments' dashboard. 2. Locate an upcoming appointment and click 'Reschedule'. 3. Verify that the system redirects to the visual calendar selection interface. 4. Select a new available date and time block from the calendar. 5. Click 'Confirm Reschedule'.	

Post Conditions: The appointment is updated with the new date/time, the previous time slot is freed in the system, and confirmation is shown.

Table 2.5 : Testing appointment rescheduling

Figure 2.19 : Successfully rescheduled appointment

2.4.2.5 Feedback and Rating System (E05)

Test Case ID: TC_FM_001	Test Designed by:
Test Title: Customer Submit New Feedback Comment	Test Designed Date:
Test Priority: High	Test Executed by:
Module Name: Feedback and Rating System	Test Executed Date:
Description: Validate that an authenticated user can create new feedback comments about an assigned staff and the respective service. This comment should create a new feedback record without altering any other feedback comments present within the system.	
Preconditions (if there are any): At least one customer account exists with ACTIVE status. At least one staff member and one service style exist in the system.	
Dependencies (if there are any): - Customer login must be functional. The feedback submission and data storage functions must be operational.	
Test Steps: 1. Log into the system by providing Customer details	

2. Navigate to the Feedback page from the customer sidebar menu.
3. Fill in the feedback comment box with a proper name and comment, choose one of the staff members from the list and pick one service from the service list.
4. Click the Submit button to submit the feedback.
5. Verify the page redirects back to the Feedback page and the newly submitted entry appears in the customer's own feedback list.
6. Ensure that the new feedback has its fields displayed correctly – name, comment, picked service, picked staff member, and the date of feedback submission.

Post Conditions: The new feedback is saved as a distinct record in the system. Previously submitted feedback entries by any customer remain unchanged and are not affected by the new submission.

Table 2.6 : Testing customer submitting a feedback

The image shows a feedback submission form and a list of feedback entries. The form has the following fields:

- NAME ***: A text input field with the placeholder "Your name".
- SERVICE RECEIVED**: A dropdown menu with "Any Service (General)" selected.
- STAFF MEMBER**: A dropdown menu with "Any Staff (General)" selected.
- COMMENT ***: A text area with the placeholder "Share your experience...".
- POST COMMENT**: A red button to submit the feedback.

Below the form, a feedback entry is displayed:

- Name**: Kavini Silva
- Date**: 2026-04-24
- Service**: CLASSIC MANICURE
- Staff**: HASINI GUNewardane
- Comment**: I went for a manicure session and was impressed with the quality of service. The staff paid close attention to hygiene and detail, which I really appreciated. The results were neat and long-lasting. The pricing is slightly on the higher side, but considering the quality and professionalism, it's worth it. I would definitely visit again and try some of their other packages as well.
- Actions**: EDIT and DELETE buttons.

Figure 2.20 : Successfully added feedback

2.4.2.6 Style Recommendation System (E06)

Test Case ID: TC_SRM_001	Test Designed by:
Test Title: Match Style to Facial Features	Test Designed Date:
Test Priority: High	Test Executed by:
Module Name: Style Recommendation System	Test Executed Date:
Description: Verify that the recommendation engine accurately provides style suggestions that best fit a customer's specific facial features and complexion.	
Preconditions (if there are any): Professional rules and logic mapping features to styles are configured in the backend.	
Dependencies (if there are any): -	
Test Steps: 1. Navigate to the Style Recommendation page. 2. Select options that match specific facial features (e.g., Round face, Oval face). 3. Select the skin tone/complexion. 4. Click the 'Get Recommendations' button.	
Post Conditions: The system successfully outputs a curated list of hairstyles and makeup looks tailored to the selected parameters.	

Table 2.7 : Test to check recommendations generated correctly

PERSONALISED FOR YOU

Discover Your *Perfect Style*

Tell us about your unique features and we will craft personalised hair and style recommendations just for you.

HAIR TYPE *

Wavy

FACE SHAPE *

Oval

EYE COLOR *

Brown

SKIN TONE *

Medium

STYLE PREFERENCE *

Natural

GET RECOMMENDATIONS

Figure 2.21 : Select inputs successfully

Your *Recommendations*

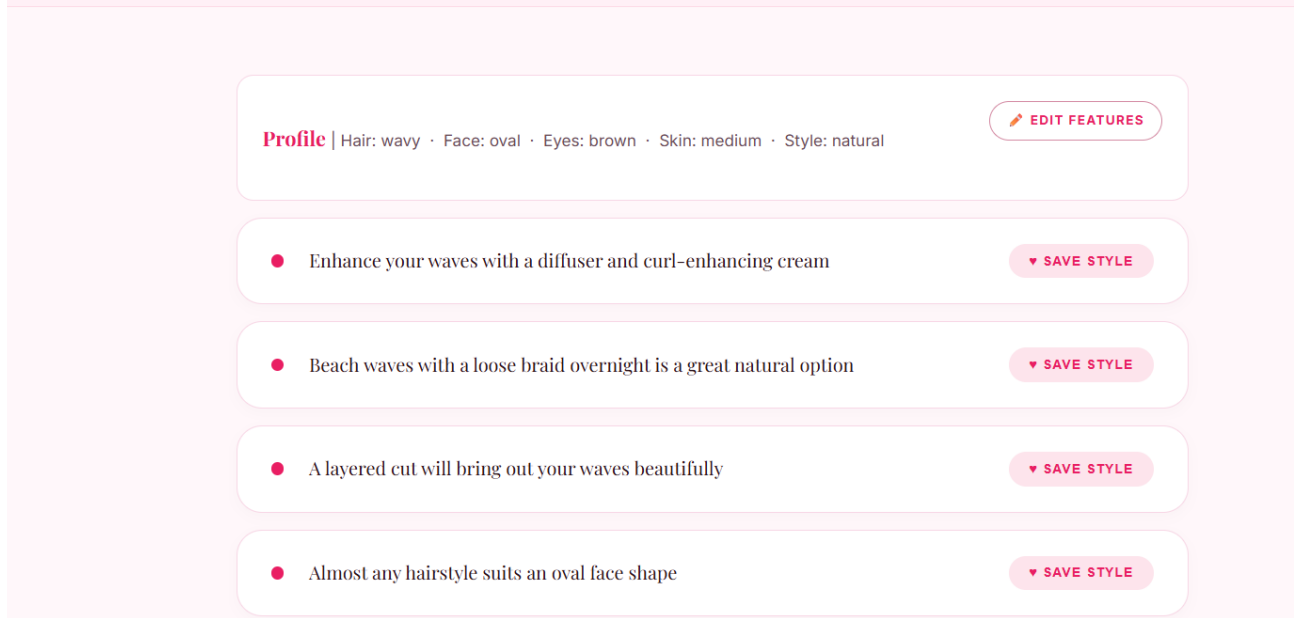


Figure 2.22 : Generated recommendations

2.4.3 Test Results Summary

Test Case ID	Module	Expected Result	Actual Result	Status
TC_UM_001	Admin and Customer Management	Customer account created successfully	Customer account created successfully	Pass
TC_SM_001	Service Management	Service added to cart correctly	Service added to cart correctly	Pass
TC_PM_001	Package Management	Package added to cart successfully	Package added to cart successfully	Pass
TC_APM_001	Appointment & Payment Management	Appointment rescheduled successfully	Appointment rescheduled successfully	Pass
TC_FM_001	Feedback & Rating System	Submitting new feedback comment	Feedback displayed correctly	Pass
TC_SRM_001	Style	Recommendations	Recommendations	Pass

	Recommendation System	generated correctly	generated correctly	
--	--------------------------	---------------------	---------------------	--

2.4.4 Testing Conclusion

Based on the testing outcomes, all the modules in the system performed as required. All the functions implemented facilitated the process of user authentication, provision of services and packages, scheduling appointments, payments, managing customer feedback, and providing style recommendations. There were no serious bugs encountered during the testing process.

3. Evaluation

3.1 Assessment of the Project results

3.1.1 Techniques Used to Analyze Results

Functional testing was employed as the main methodology of assessing the system by checking that the outputs produced by each module coincided with what they should have done according to the stated requirements. Test scenarios were generated for each epic and included normal flows, validations, and boundary conditions. Black-box testing of each of the system roles (Owner, Receptionist, Staff, and Customer) was performed manually.

Also, user acceptance criteria specified during the requirements gathering stage were used as criteria to assess the quality of developed functionality in terms of how well it satisfied its intended use and fit into the corresponding parts of the system.

3.1.2 Results of the Analysis

From the results of the evaluation, the main goal of designing the web-based salon management application system has been realized. This is because the developed modules include functionality for the management of customers, services and packages, scheduling of appointments, handling of payments, providing feedback and giving recommendations for styles. All these functionalities solve the identified challenges in the analysis phase which include keeping records manually, scheduling conflicting appointments, and difficulty in managing customers' information.

The core modules have been implemented efficiently and according to the requirements. First, the appointment booking module ensures that there are no double bookings by blocking out the unavailable slots. Secondly, the payment module verifies payment information and masks sensitive information such as the card details. Thirdly, the style recommendation module generates personalized recommendations from the customers' characteristics and preferences. Fourthly, access control is used to ensure that users access only the appropriate pages based on their roles. Finally, the feedback and rating module allows customers to add, edit, and delete their feedbacks and the best feedback can even appear on the public highlights page.

3.1.3 Limitations and Shortcomings

While the objectives have been met despite certain challenges in their achievement, some limitations have also emerged upon the assessment.

The payment processing module merely imitates the verification process instead of connecting to an actual payment gateway. Although the payment information is verified, the payment process itself is not performed. As such, the module can only be used for demo purposes and should not be implemented into practice.

The style recommendation module relies on fixed business logic rather than using any AI or machine learning tools. Recommendations will not become more accurate with time as they are based on the same principles.

The customer purchase history is based on appointment records rather than transactions. Therefore, cancelled bookings may show up in this record even though there was no actual purchase and users have to evaluate their appointment status accordingly.

Moreover, the application lacks notifications via email or SMS about confirmed bookings and cancellations. The application does not provide the owner with business statistics such as customer retention and popular services.

3.1.4 Identified Deficiencies and Possible Improvements

Several possible enhancements have been identified through the evaluation process. Firstly, the lack of a notification feature diminishes user convenience since users will have to manually log into the system just to confirm whether their bookings have been updated. Integrating features such as automated emails and/or SMS notifications will enhance communication between the company and its customers.

In addition, the current reporting function can be enhanced by adding business intelligence dashboards that go beyond monthly financial summaries to incorporate statistics on services offered, bookings made and even customer behavior analysis.

Moreover, the feedback management feature can be enhanced by introducing a moderation function that enables administrators to filter out user feedback prior to making it live. Additionally, the introduction of stronger data input validation can prevent any issues associated with malicious input.

3.2 Lessons Learned

3.2.1 Planning and Requirement Definition

Early on, we realized how critical it is to nail down system requirements and map out data relationships before jumping into development. What seemed like simple features at first often forced us to tweak the data model later on. It would've saved us a lot of backtracking if we'd spent more time on analysis and setting up the database structure right from the start.

3.2.2 Team Collaboration and Integration

Since the project was developed in collaboration by different developers, there arose challenges when trying to combine individual systems together. There were naming and implementation discrepancies which posed challenges in the integration of the individual components. This highlighted the importance of having coding standards and development guidelines in place to ensure easy integration.

3.2.3 Security and Access Control

The application of role-based access control proved the importance of ensuring security in application development at the initial stages of the project. Security concerns emerged during testing as a number of the access restrictions had to be adjusted so that the user was limited to accessing certain authorized functionalities.

3.2.4 Scope Management

It is important to highlight the importance of scope management in project development. In this project, there arose a need for additional features, such as online payment gateways, notifications,

and analytics tools. However, it was realized that these features would take too much time and would be difficult to develop. Therefore, it is vital to limit development and ensure that core functionalities are achieved first.

3.2.5 Testing and Quality Assurance

The team came to realize that testing needs to occur at regular intervals during development, rather than towards the end of the project. Some problems arose towards the later stages of development and were consequently harder to fix, due to their potential interaction with already established code.

3.3 Future Work

Even though the developed system meets the existing project requirements, there are various possible improvements that can be suggested for its future versions. One of them is to implement the real online payment gateway, such as Koko, MintPay or PayPal, so that the customers would be able to make real payments and get their receipt and payment confirmation. The introduction of automated email and SMS notifications would make appointments easier for the customers as well as provide confirmation, reminder and even cancellation notification. This would ensure that there were fewer missed appointments. There is currently an existing rule-based recommendation module which can use artificial intelligence and machine learning to make personalized style recommendations for the customers. The more accurate recommendations would be generated by the system. One possible future enhancement of the current project would be introducing analytics and reports that can help determine service popularity, high demand times, employee performance, retention statistics and revenue trends. This would allow making better decisions based on available information and contribute to growing the business.

Other possible enhancements might involve developing a loyalty rewards program where customers would earn points by booking services and exchanging them for discounts or other offers. The introduction of a mobile app would improve the experience of both the customers and the salon employees in terms of managing bookings and appointments as well as receiving notifications.

Finally, it is important to have a feedback moderation function that will help avoid any inappropriate comments being published.

4. Conclusion

The purpose of developing the Salon Harshi Management System was to automate manual salon procedures by introducing a centralized digital system that could serve as an efficient tool for managing Owners, Receptionists, Staff, and Customers. The development process has succeeded in fulfilling this purpose through the introduction of functionalities for managing users, services and packages, appointment bookings, payments, style recommendations, and customer feedback.

Introducing the developed application will bring several positive outcomes for the Harshi Salon. First, online booking will help avoid double bookings and facilitate managing appointments. Centralized management of customers' and service-related information will ensure the accuracy and availability of data records. Additionally, the feedback and ratings system will provide the owner with insights into customers' opinions, while the style recommendations functionality will improve customers' experience.

Although the requirements of the project have been fulfilled, there are some constraints. The payment module is only a simulation of the online payment gateway system rather than an actual payment gateway. Additionally, the style recommendation engine operates using predefined rules rather than considering customer behavior and preferences. The absence of automation, such as notifications and advanced business analytics, could also be considered a constraint.

All these constraints could be mitigated through future enhancements to the system, such as the integration of the payment gateway, implementing machine learning-based recommendation techniques, and adding the automatic sending of e-mails or SMS messages. These enhancements will make the system more scalable and useful.

In conclusion, digital transformation is shown to be a useful process for managing a small service-oriented business. The system created in the project represents a good basis for future development and implementation.

5. References

[1] diagrams.net, “draw.io,” 2024. [Online]. Available: <https://www.diagrams.net/>. [Accessed: 24-May-2026].

[2] Lucid Software Inc., “Lucidchart,” 2024. [Online]. Available: <https://www.lucidchart.com/>. [Accessed: 24-May-2026].

[3] Visual Paradigm International Ltd., “Visual Paradigm Online,” 2024. [Online]. Available: <https://www.visual-paradigm.com/>. [Accessed: 24-May-2026].

[4] Figma Inc., “Figma: Collaborative Interface Design Tool,” 2024. [Online]. Available: <https://www.figma.com/>. [Accessed: 24-May-2026].

[5] Miro, “Miro,” 2024. [Online]. Available: <https://miro.com/>. [Accessed: 24-May-2026].

[6] Atlassian, “Jira Software,” 2024. [Online]. Available: <https://www.atlassian.com/software/jira>. [Accessed: 24-May-2026].

[7] Sri Lanka Institute of Information Technology (SLIIT), “IE2021 Requirement Engineering – Assessment 02 report,” Internal Course Material, 2024.

[8] Sri Lanka Institute of Information Technology (SLIIT), “IE2091 Sprint 0 Document,” Internal Course Material, 2024.

Appendix A: Design Diagrams

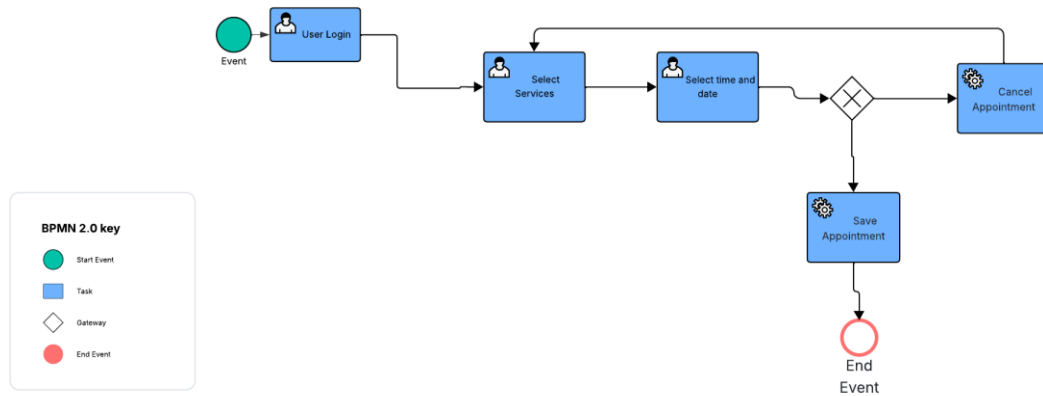


Figure 5.1 : Book appointment process model

Appendix B: Test Results

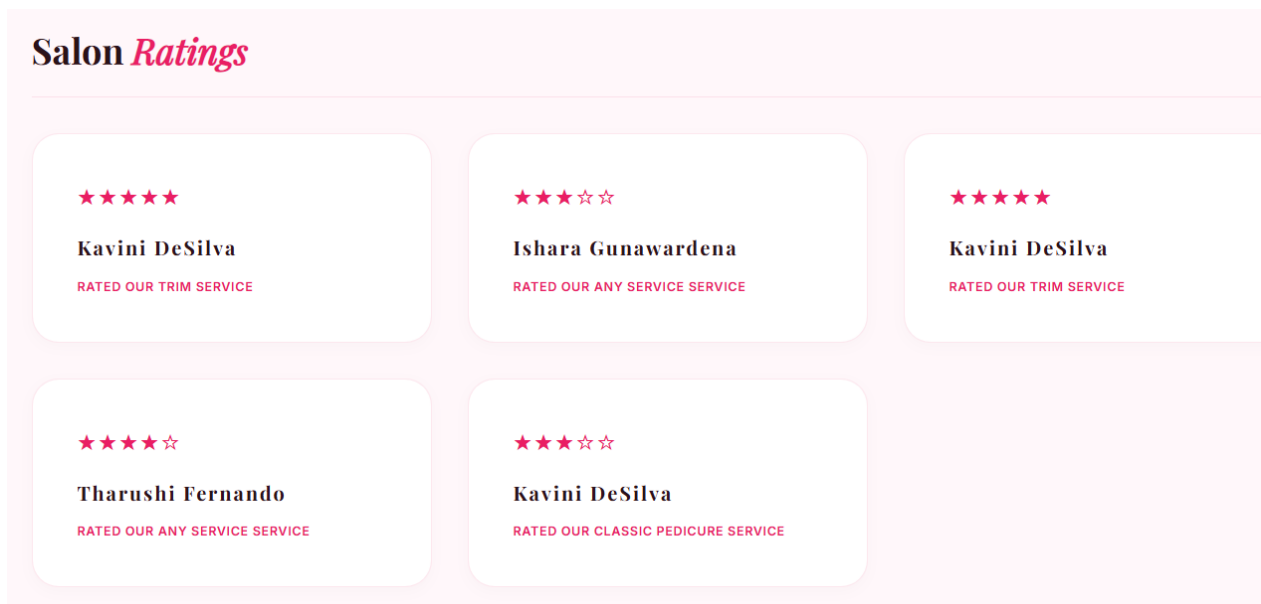


Figure 5.2 : Display of customer ratings

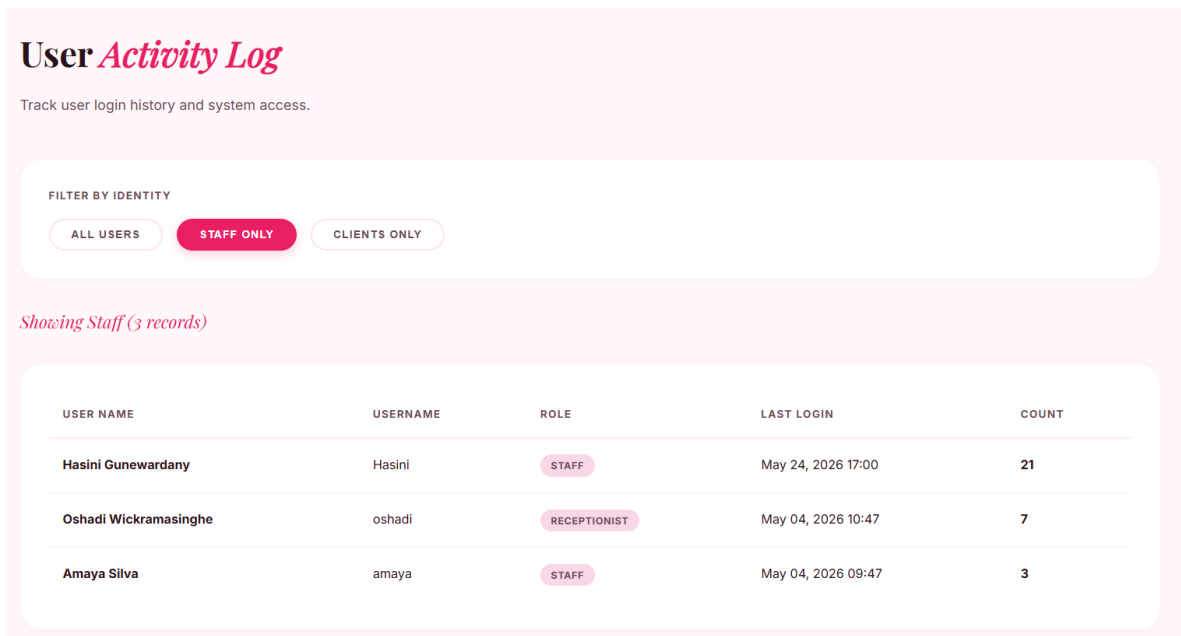


Figure 5.3 : Admin can view all users and their login count

Appendix C: Selected Code Listings

```
@Bean
public AuthenticationSuccessHandler authenticationSuccessHandler() {
    return (HttpServletRequest request, HttpServletResponse response, Authentication authentication) -> {
        String role = authentication.getAuthorities().stream()
            .map(capture of extends GrantedAuthority grantedAuthority -> grantedAuthority.getAuthority())
            .filter(String auth -> auth.startsWith("ROLE_"))
            .map(String auth -> auth.substring(beginIndex: 5))
            .findFirst()
            .orElse(other: "CUSTOMER");

        String redirectUrl = switch (role) {
            case "OWNER" -> "/owner/dashboard";
            case "RECEPTIONIST" -> "/reception/dashboard";
            case "STAFF" -> "/staff/dashboard";
            default -> "/customer/dashboard";
        };

        response.sendRedirect(redirectUrl);
    };
}
```

Figure 5.4 : Role-Based Authentication and Login Redirect

```
@Query("SELECT new org.example.dto.MonthlyRevenueDTO(MONTH(a.appointmentDate), SUM(a.totalPrice)) " + 1 usage
"FROM Appointment a " +
"WHERE YEAR(a.appointmentDate) = :year AND a.status = :status " +
"GROUP BY MONTH(a.appointmentDate) " +
"ORDER BY MONTH(a.appointmentDate)")
List<MonthlyRevenueDTO> getMonthlyRevenueForYear(@Param("year") int year, @Param("status") AppointmentStatus status);
```

Figure 5.5 : Monthly Revenue Query

```

public class AppointmentService {

    public Appointment createAppointment(AppointmentRequest req, Long customerId) { 1usage
        if (getBookedTimes(req.appointmentDate).contains(req.appointmentTime)) {
            throw new RuntimeException("This time slot is already booked. Please select another.");
        }

        Appointment appt = new Appointment();
        appt.setCustomerName(req.customerName);
        appt.setCustomerPhone(req.customerPhone);
        appt.setAppointmentDate(req.appointmentDate);
        appt.setAppointmentTime(req.appointmentTime);
        appt.setTotalPrice(req.totalPrice);
        appt.setBookedItemsJson(req.bookedItemsJson);
        appt.setStatus(AppointmentStatus.PENDING);

        // NEW: Link the logged-in customer to the appointment
        if (customerId != null) {
            Customer customer = customerRepository.findById(customerId)
                .orElseThrow(() -> new RuntimeException("Customer not found"));
            appt.setCustomer(customer);
        }

        // Link the staff member
        if (req.staffId != null) {
            Employee staff = employeeRepository.findById(req.staffId)
                .orElseThrow(() -> new RuntimeException("Stylist not found"));
            appt.setAssignedStaff(staff);
        }

        return repository.save(appt);
    }
}

```

Figure 5.6 : Appointment Booking with Time Slot Conflict Prevention

```

// 3. Process Mock Payment with Validations
public Appointment processPayment(Long id, PaymentRequest payment) { 1usage
    Appointment appt = repository.findById(id)
        .orElseThrow(() -> new RuntimeException("Appointment not found"));

    // Validations
    if (payment.cardNumber == null || !payment.cardNumber.matches(regex: "\\d{16}")) {
        throw new RuntimeException("Invalid Card Number: Must be 16 digits.");
    }

    if (payment.expiryDate == null || !payment.expiryDate.matches(regex: "(0[1-9]|1[0-2])\\/(\\d{2})")) {
        throw new RuntimeException("Invalid Expiry Date: Must be MM/YY format.");
    }

    if (payment.cvv == null || !payment.cvv.matches(regex: "\\d{3}")) {
        throw new RuntimeException("Invalid CVV: Must be 3 digits.");
    }

    // Mask the card for history view (e.g., **** * 1234)
    String masked = "**** * " + payment.cardNumber.substring(beginIndex: 12);
    appt.setMaskedCardNumber(masked);
    appt.setStatus(AppointmentStatus.PAID);

    return repository.save(appt);
}

```

Figure 5.7 : Payment Validation and Card Masking

```
// — Recommendation logic —————
private List<String> buildRecommendationTexts(StyleRequest request) { 2 usages
    List<String> recs = new ArrayList<>();

    if (request.getHairType() != null) {
        switch (request.getHairType()) {
            case "straight":
                recs.add("Try a sleek blowout or flat iron style for a polished look");
                recs.add("Side-swept bangs work beautifully with straight hair");
                recs.add("Consider a lob (long bob) cut for easy styling");
                break;
            case "wavy":
                recs.add("Enhance your waves with a diffuser and curl-enhancing cream");
                recs.add("Beach waves with a loose braid overnight is a great natural option");
                recs.add("A layered cut will bring out your waves beautifully");
                break;
            case "curly":
                recs.add("Use the LOC method (Leave-in, Oil, Cream) to define your curls");
                recs.add("A DevaCut from a curly hair specialist is highly recommended");
                recs.add("Avoid brushing dry – detangle only with conditioner");
                break;
            case "coily":
                recs.add("Deep conditioning weekly will keep coily hair moisturized");
                recs.add("Protective styles like twists or braids reduce breakage");
                recs.add("Try the pineapple method at night to preserve your style");
                break;
        }
    }
}
```

Figure 5.8 : Style Recommendation Algorithm

```
if (request.getFaceShape() != null) {
    switch (request.getFaceShape()) {
        case "oval":
            recs.add("Almost any hairstyle suits an oval face shape");
            recs.add("Try a curtain bang to frame your balanced features");
            break;
        case "round":
            recs.add("Long layers and side parts help elongate a round face");
            recs.add("High updos add height and slim the face nicely");
            break;
        case "square":
            recs.add("Soft waves and curls soften a strong jawline");
            recs.add("Side-swept styles and wispy bangs complement a square face");
            break;
        case "heart":
            recs.add("Chin-length bobs and lobes balance a wider forehead");
            recs.add("Add volume at the jaw area to balance heart-shaped features");
            break;
    }
}
```

Figure 5.9 : Style Recommendation Algorithm

```

if (request.getSkinTone() != null) {
    switch (request.getSkinTone()) {
        case "fair":
            recs.add("Cool ash or platinum blonde tones complement fair skin beautifully");
            recs.add("Soft pink or rose gold highlights look stunning on fair skin");
            break;
        case "medium":
            recs.add("Warm caramel and honey highlights flatter medium skin tones");
            recs.add("Copper and auburn tones add a gorgeous glow to medium skin");
            break;
        case "dark":
            recs.add("Bold colors like burgundy and deep violet look stunning on dark skin");
            recs.add("Warm honey highlights create a beautiful contrast on dark skin");
            break;
        case "olive":
            recs.add("Golden and warm brown tones harmonize perfectly with olive skin");
            recs.add("Warm ombre from dark roots to caramel ends suits olive tones");
            break;
    }
}

```

Figure 5.10 : Style Recommendation Algorithm

```

if (request.getStylePreference() != null) {
    switch (request.getStylePreference()) {
        case "classic":
            recs.add("A timeless blowout or chignon never goes out of style");
            recs.add("Structured updos and neat ponytails are always elegant");
            break;
        case "modern":
            recs.add("Try a textured lob or an asymmetric cut for a modern edge");
            recs.add("Sleek straight styles with middle parts are very contemporary");
            break;
        case "bold":
            recs.add("Go for a bold color like vivid red, electric blue, or pastel pink");
            recs.add("Try an avant-garde updo with unexpected texture or volume");
            break;
        case "natural":
            recs.add("Embrace your natural texture with minimal heat styling");
            recs.add("Try protective styles that celebrate your natural pattern");
            break;
    }
}

return recs;
}

```

Figure 5.11 : Style Recommendation Algorithm